

Walk-forward performance (replay)

Fold × strategy × symbol from `backtest_executor.py --replay --include=trades` . **Expectancy** = mean PnL per trade; **PF** = $\text{sum}(\text{winning \$}) / |\text{sum}(\text{losing \$})|$; **Max DD** = peak-to-trough on cumulative PnL within that fold’s trade sequence; **ΣR** = $\text{sum of } (\text{trade PnL} / \text{initial_risk_dollars})$ where risk was recorded.

`metrics.json` · `metrics_insights.json`

All folds pooled: 18 trades · PnL **1448.50** · expectancy **80.472** / trade · σ **152.911** · PF **3.63** · max consecutive losses **2** · max DD (sequential) **148.50**

Survivability read: high *max consecutive losses* inside a fold with negative expectancy usually means the edge is not stable across that window. Compare the *rollup* table (all folds merged per strategy×symbol) to per-fold rows — widening σ or falling PF in later folds is a simple walk-forward stress signal.

Per fold

fold	start	end	strategy	sym	n	PnL	WR%	E/trade	σ PnL	PF	maxDD	win	loss	max win str	max loss str	avg bars	ΣR
0	2026-04-26	2026-04-28	morning_range_reversion	MNQ	1	333.25	100.0%	333.250	0.000	—	0.00	1	0	1	0	17.0	1.74
1	2026-04-29	2026-05-01	morning_range_reversion	MNQ	2	155.50	50.0%	77.750	193.040	3.65	58.75	1	1	1	1	7.0	0.61
2	2026-05-02	2026-05-04	morning_range_reversion	MNQ	1	-75.25	0.0%	-75.250	0.000	0.00	75.25	0	1	0	1	2.0	-1.08
3	2026-05-05	2026-05-07	morning_range_reversion	MNQ	2	154.50	50.0%	77.250	185.969	3.85	54.25	1	1	1	1	26.5	0.59
4	2026-05-08	2026-05-10	morning_range_reversion	MNQ	1	84.75	100.0%	84.750	0.000	—	0.00	1	0	1	0	3.0	1.59
5	2026-05-11	2026-05-13	morning_range_reversion	MNQ	3	372.75	100.0%	124.250	40.814	—	0.00	3	0	3	0	5.7	4.92
6	2026-05-14	2026-05-16	morning_range_reversion	MNQ	2	8.00	50.0%	4.000	160.867	1.07	109.75	1	1	1	1	1.0	0.59
7	2026-05-17	2026-05-19	morning_range_reversion	MNQ	2	294.00	50.0%	147.000	356.735	3.79	105.25	1	1	1	1	6.0	0.70
8	2026-05-20	2026-05-22	morning_range_reversion	MNQ	3	185.75	66.7%	61.917	127.574	3.22	83.75	2	1	2	1	6.7	2.25
9	2026-05-23	2026-05-26	morning_range_reversion	MNQ	1	-64.75	0.0%	-64.750	0.000	0.00	64.75	0	1	0	1	2.0	-1.09

Rollup — all folds merged (per strategy × symbol)

strategy	sym	n	PnL	WR%	E/trade	σ PnL	PF	maxDD	max loss str	avg bars	ΣR
morning_range_reversion	MNQ	18	1448.50	61.1%	80.472	152.911	3.63	148.50	2	7.9	10.82

Simulated account equity

Starting balance **\$2,000.00**. Every completed trade in this run is sorted by **exit time** and applied in sequence to one notional account (all legs, all folds). This shows combined PnL path stress; it does *not* model margin, cross-margin, or whether every leg would run on the same live account.

Equity path: final **3,448.50** · return **72.42%** · max drawdown **148.50 (4.13%** from peak) · closes in path **18**



Analysis — all legs pooled

All legs pooled (exit-time order, all folds)

Performance insights

Metric	Value	Note
Breakeven win rate (from avg win / avg loss)	30.2%	Actual WR 61.1%
Recovery factor (total PnL / sequential max DD)	9.754	Higher is better; uses fold-style cumulative PnL DD
Median bars held	W:10.0 / L:2.0	Win vs loss holding time
Avg R multiples	win:1.672 loss:-1.081	Where initial_risk_dollars recorded

Losses by exit reason

exit_reason	% of losses	detail
stop_loss	100.0%	losses 7 / events 7

Loss rate by side

side	Loss rate	detail
BUY	33.3%	3 losses / 9 trades
SELL	44.4%	4 losses / 9 trades

Loss mix by strategy leg

Populated when trades carry `_strategy` from the recap report (multi-strategy runs).

strategy	Losses	Context
morning_range_reversion	7	100.0% of all losses · 18 trades in leg

Loss count by entry weekday (ET)

Weekday	Losses	Context
Mon	1	14.3% of all losses; 3 entries this weekday
Tue	2	28.6% of all losses; 4 entries this weekday
Wed	1	14.3% of all losses; 4 entries this weekday
Thu	1	14.3% of all losses; 4 entries this weekday
Fri	2	28.6% of all losses; 3 entries this weekday

Linear association with losses (exploratory)

Pearson correlation between a 0/1 *loss indicator* and each factor. Values near 0 mean no obvious linear pattern in this sample; $|r|>0.15$ is a loose rule-of-thumb for “worth eyeballing” only.

Factor	r vs loss	Interpretation
bars_held	-0.4032	When bars_held is lower, losing trades are slightly more common (Pearson $r\approx-0.40$); treat as exploratory—causation not implied.
initial_risk_dollars	-0.3512	When initial_risk_dollars is lower, losing trades are slightly more common (Pearson $r\approx-0.35$); treat as exploratory—causation not implied.
entry_hour_et	0.1849	When entry_hour_et is higher, losing trades are slightly more common (Pearson $r\approx0.18$); treat as exploratory—causation not implied.

Analysis — per strategy × symbol (merged folds)

morning_range_reversion · MNQ (merged folds)

Performance insights

Metric	Value	Note
Breakeven win rate (from avg win / avg loss)	30.2%	Actual WR 61.1%
Recovery factor (total PnL / sequential max DD)	9.754	Higher is better; uses fold-style cumulative PnL DD
Median bars held	W:10.0 / L:2.0	Win vs loss holding time
Avg R multiples	win:1.672 loss:-1.081	Where initial_risk_dollars recorded

Losses by exit reason

exit_reason	% of losses	detail
stop_loss	100.0%	losses 7 / events 7

Loss rate by side

side	Loss rate	detail
BUY	33.3%	3 losses / 9 trades
SELL	44.4%	4 losses / 9 trades

Loss mix by strategy leg

Populated when trades carry `_strategy` from the recap report (multi-strategy runs).

strategy	Losses	Context
morning_range_reversion	7	100.0% of all losses · 18 trades in leg

Loss count by entry weekday (ET)

Weekday	Losses	Context
Mon	1	14.3% of all losses; 3 entries this weekday
Tue	2	28.6% of all losses; 4 entries this weekday
Wed	1	14.3% of all losses; 4 entries this weekday
Thu	1	14.3% of all losses; 4 entries this weekday
Fri	2	28.6% of all losses; 3 entries this weekday

Linear association with losses (exploratory)

Pearson correlation between a 0/1 *loss indicator* and each factor. Values near 0 mean no obvious linear pattern in this sample; $|r|>0.15$ is a loose rule-of-thumb for “worth eyeballing” only.

Factor	r vs loss	Interpretation
bars_held	-0.4032	When bars_held is lower, losing trades are slightly more common (Pearson $r\approx-0.40$); treat as exploratory—causation not implied.
initial_risk_dollars	-0.3512	When initial_risk_dollars is lower, losing trades are slightly more common (Pearson $r\approx-0.35$); treat as exploratory—causation not implied.
entry_hour_et	0.1849	When entry_hour_et is higher, losing trades are slightly more common (Pearson $r\approx0.18$); treat as exploratory—causation not implied.